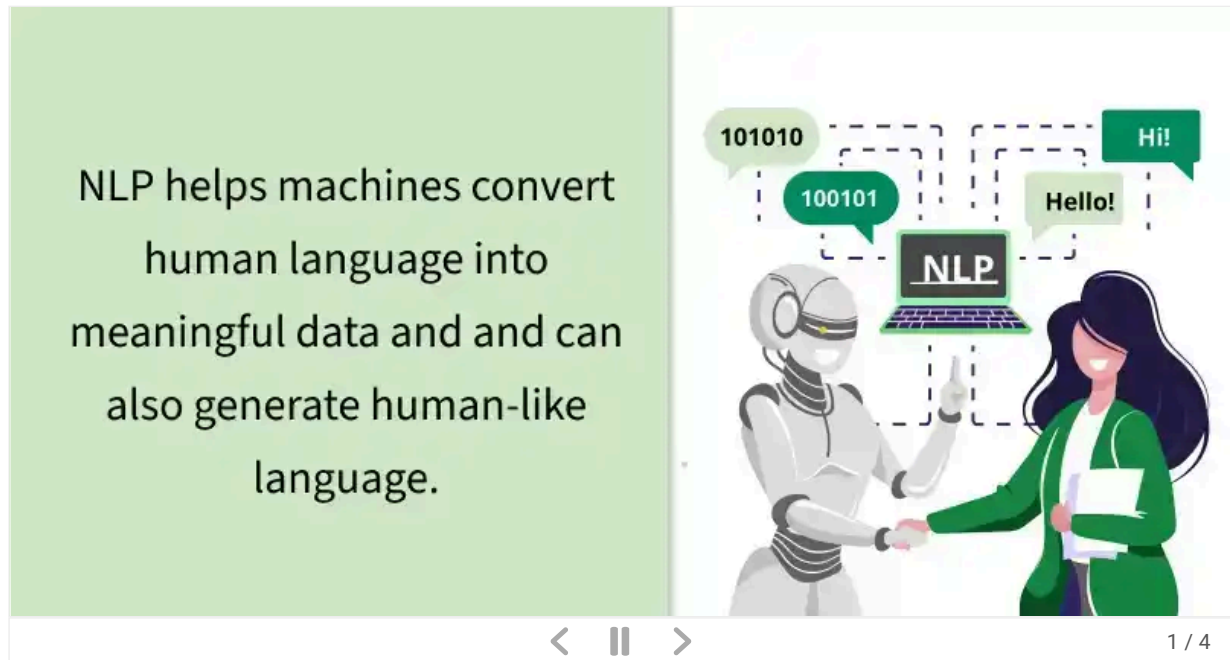


Introduction to Natural Language Processing (NLP)

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Natural Language Processing (NLP) helps computers understand, interpret and produce human language. It studies language as data and develops a model that can analyse linguistic structure, meaning and context in both written and spoken communication.



Simple Example of NLP: **"Ravi is happy with the new phone."**

An NLP system can:

- Detect Ravi as a person
- Identify phone as an object
- Recognize sentiment as positive
- Understand topic as product review

How Natural Language Processing Works

1. Text or Speech Input

- **Receiving text data:** The system takes written language like sentences or documents which is called text acquisition.
- **Receiving voice input:** When the input is audio, it is first converted into text using Speech Recognition.

2. Pre-processing

The text is cleaned and prepared. It can include:

- **Removing punctuation or noise:** Cleaning unwanted characters or symbols from text is done using text normalization.
- **Splitting into words:** Breaking sentences into smaller units so they can be processed easily.
- **Converting to lowercase:** Changing all words into the same case for uniform processing is known as case folding.
- **Removing common words:** Eliminating frequent words like is, the, and to focus on meaningful terms.
- **Reducing words to base form:** Converting words like running to run to reduce computational power.

3. Language Analysis

The system studies structure and meaning:

- **Grammar detection:** Identifying nouns, verbs, and other parts of speech in a sentence is done.
- **Word relationships:** Finding how words connect to each other in a sentence.
- **Context understanding:** Determining the actual meaning of a word based on surrounding text.
- **Finding names and places:** Detecting entities like person names, locations, or dates.
- **Sentiment detection:** Identifying whether text expresses positive, negative or neutral emotion.

4. Text Representation and Embedding Techniques

Since machines process numbers, this stage converts text into numerical vectors.

- **Text representation:** In this step, text is converted into numbers using statistical features or vector representations so machines can process it.
- **Traditional representations:** Earlier methods represent text using word counts and importance scores.
- **Word embeddings:** Modern methods represent words as dense vectors capturing similarity and meaning.
- **Contextual embeddings:** Advanced models generate word meanings based on the surrounding sentence.

5. Model Training

Once text is numeric, models are trained to learn patterns and perform NLP tasks.

- **Model training:** After text is converted into vectors, algorithms learn patterns from data to perform tasks like classification or translation.
- **Traditional machine learning:** Earlier NLP systems relied on statistical algorithms that learn from manually prepared features.
- **Deep learning approaches:** Modern NLP uses neural networks that automatically learn language structure from large data.
- **Pre-trained models:** Large language models trained on massive datasets can be reused and fine-tuned for tasks.

6. Output Generation

The system produces results such as:

- Text reply

- Voice response
- Translation
- Summary
- Prediction

Common NLP Tasks

- **Text classification:** Assigning predefined labels to text like spam or topic categories.
- **Sentiment analysis:** Detecting whether text expresses positive, negative or neutral emotion.
- **Machine translation:** Automatically converting text from one language to another.
- **Named Entity Recognition:** Identifying names of people, places, dates, etc in text.
- **Text summarization:** Generating a shorter version of a document while keeping key meanings.
- **Question answering systems:** Systems that read text and return exact answers to queries.

Real-Life Applications

- Voice assistants like Alexa, Google Assistant, etc
- Chatbots in customer support
- Email spam filtering
- Auto-correct and predictive typing
- Language translation tools
- Social media sentiment tracking
- Document search and recommendation systems

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